

EXTREME RAYS OF THE CONE OF LOG-CONCAVITY

CLAUDE (FABLE-5.1), WITH CHATGPT-6 (ASTRA)

ABSTRACT. White conjectured that a product of Schur functions indexed by partitions with at most two parts spans an extreme ray of the cone of log-concavity if and only if its factors form a nested multiset. We prove the conjecture. The main ingredient is a specialization of the complete homogeneous symmetric functions to a positive log-concave sequence. Making its logarithmic increments constant on an interval, and then perturbing them, turns a positive decomposition of Schur products into an identity between products of linear forms. Interval counts constrain the endpoints of these forms. A suitable coefficient then singles out the original nested multiset, with dominance supplying the final comparison when a one-part factor is present.

1. INTRODUCTION

Let $\Lambda = \mathbb{R}[h_1, h_2, \dots]$ be the ring of symmetric functions, with $\deg h_i = i$ and $h_0 = 1$. For $N \geq 1$, let $\mathcal{SP}_N$ be the set of multisets of nonempty partitions with at most two parts and total size N . For $A \in \mathcal{SP}_N$, write

$$s_A = \prod_{\rho \in A} s_\rho, \quad C_N = \text{cone}\{s_A : A \in \mathcal{SP}_N\} \subseteq \Lambda_N.$$

Jacobi–Trudi gives

$$(1) \quad s_{(a,b)} = h_a h_b - h_{a+1} h_{b-1} \quad (a \geq b \geq 1), \quad s_{(c)} = h_c.$$

Thus C_N is the degree- N part of White's *cone of log-concavity*. Its generators are nonnegative on every positive log-concave sequence $(h_i)_{i \geq 0}$. The question is which of these generators are necessary: which products span extreme rays?

We call a two-part member (a, b) of A a *pair*, with *top* a and *bottom* b , and a one-part member (c) a *singleton*. Following White, two members *interlace* if, after possibly exchanging them, they are one of the following:

- (1) pairs (a, b) and (d, e) with $a > d \geq b > e$;
- (2) a pair (a, b) and a singleton (c) with $a > b$ and $a \geq c \geq b$;
- (3) two singletons.

A multiset is *nested* if no two of its members interlace. Repeated members are counted with multiplicity. In particular, a pair (p, p) interlaces with nothing, and a nested multiset has at most one singleton.

THEOREM 1.1. *For every $N \geq 1$ and $A \in \mathcal{SP}_N$, the product s_A spans an extreme ray of C_N if and only if A is nested.*

This proves White's Conjecture 3, listed as OPAC-009 [3]. White proved the necessity of nestedness and established the converse when all the parts of the members of A are distinct [1, Theorems 4 and 12]. Our proof allows arbitrary repetitions.

Here is the idea. A specialization $h_k \mapsto g(k) > 0$ with constant logarithmic increments on $[x, y + 1]$ kills exactly the factors $s_{(a,b)}$ with $x \leq b \leq a \leq y$. A perturbation of these increments replaces each vanishing factor, to first order, by a positive constant times $\delta_b - \delta_{a+1}$. Positivity prevents cancellation of the lowest powers of the perturbation parameter. Consequently, any positive decomposition of s_A forces each competitor to have at least as many pairs as A in every interval.

These interval inequalities control the sorted lists of tops and bottoms. If A has no singleton, the coefficient of its all-bottom monomial finishes the proof. If A has a singleton (c) , nestedness splits its pairs into blocks below, at, and above c . We select the top endpoints below c and the bottom endpoints above c ; dominance then forces both blocks, and the singleton, to agree with A .

2. THE REDUCTION TO NESTED COMPETITORS

For a multiset A , let $\varphi(A)$ be the partition obtained by sorting all its parts. Write $\mu \succeq \lambda$ for dominance order, with zeros appended when needed. White's Schur-support statement is

$$(2) \quad s_A = \sum_{\mu \succeq \varphi(A)} c_A^\mu s_\mu, \quad c_A^\mu \geq 0, \quad c_A^{\varphi(A)} = 1 \quad [1, \text{Proposition 6}].$$

We use it together with White's theorem that non-nested products are not extreme. No result from the withdrawn paper [2] is needed.

LEMMA 2.1. *Distinct multisets in $\mathcal{SP}_N$ give nonproportional products s_A .*

Proof. The polynomial $h_a h_b - h_{a+1} h_{b-1}$ is primitive and linear in h_{a+1} over the polynomial ring in the other variables: its two coefficients are coprime. Gauss's lemma therefore makes it irreducible. The polynomials h_c are also irreducible. These irreducibles are pairwise nonassociate: a two-part factor has largest variable index $a + 1$, and its coefficient of that variable is $-h_{b-1}$ (or -1 if $b = 1$), which determines b and its normalization. A one-part factor cannot be associate to a two-term polynomial. Unique factorization now proves the assertion. All these arguments take place in a polynomial ring with finitely many variables for any given products. $\square$

LEMMA 2.2. *If A is nested and s_A does not span an extreme ray, then*

$$(3) \quad s_A = \sum_{B \in \mathcal{S}} c_B s_B, \quad c_B > 0,$$

where $\mathcal{S}$ is a nonempty finite set of nested multisets $B \neq A$, each satisfying $\varphi(B) \succeq \varphi(A)$.

Proof. The cone C_N is finitely generated and pointed, since it lies in the Schur-positive cone. It is therefore generated by its extreme rays, each of which contains a generator s_B . Such a B is nested by White's theorem and Lemma 2.1. Express s_A as a positive combination of these generators. None is proportional to s_A , since s_A is assumed not to span an extreme ray. Finally, the coefficient of $s_{\varphi(B)}$ on the right is positive. There is no cancellation in the Schur basis, so (2) forces $\varphi(B) \succeq \varphi(A)$. $\square$

3. AN INTERVAL SPECIALIZATION

For an integer interval $I = [x, y]$, with $1 \leq x \leq y$, let B_I be the multiset of pairs $(a, b) \in B$ with $x \leq b \leq a \leq y$, and set

$$n_I(B) = |B_I|, \quad P_{B_I}(\delta) = \prod_{(a,b) \in B_I} (\delta_b - \delta_{a+1}).$$

The variables of this polynomial are indexed by $T = [x, y + 1]$. An empty product is 1.

LEMMA 3.1 (Interval degeneration). *Suppose (3) holds, without requiring A or the B 's to be nested. Fix $I = [x, y]$ and put $m = n_I(A)$. Then*

$$(4) \quad n_I(B) \geq m \quad (B \in \mathcal{S}).$$

Moreover, there are constants $K_A, K_B > 0$, independent of δ , such that

$$(5) \quad K_A P_{A_I}(\delta) = \sum_{\substack{B \in \mathcal{S} \\ n_I(B) = m}} c_B K_B P_{B_I}(\delta).$$

Proof. Choose a nonincreasing sequence $d^0(k)$, $k \geq 1$, whose only equal consecutive values occur within T : it is constant on T and strictly decreases at every other step. For $e = (e_k)_{k \in T}$, extended by zero outside T , set

$$d^t(k) = d^0(k) + t e_k, \quad g_t(k) = \exp\left(\sum_{j=1}^k d^t(j)\right), \quad \psi_{t,e}(h_k) = g_t(k).$$

Here $g_t(0) = 1$, and algebraic independence of the h_k makes $\psi_{t,e}$ a well-defined ring homomorphism. By (1),

$$(6) \quad \psi_{t,e}(s_{(a,b)}) = g_t(a)g_t(b) (1 - \exp(d^t(a+1) - d^t(b))).$$

At $t = 0$ this is zero exactly when (a, b) lies inside I ; otherwise it is positive. For a pair inside I , Taylor expansion gives

$$\psi_{t,e}(s_{(a,b)}) = t g_0(a)g_0(b)(e_b - e_{a+1}) + O(t^2).$$

Multiplying the factors, for each fixed e we obtain

$$(7) \quad \psi_{t,e}(s_B) = t^{n_I(B)} K_B P_{B_I}(e) + O(t^{n_I(B)+1}),$$

where

$$K_B = \prod_{(a,b) \in B_I} g_0(a)g_0(b) \prod_{\substack{(a,b) \in B \\ (a,b) \notin B_I}} \psi_{0,e}(s_{(a,b)}) \prod_{(c) \in B} g_0(c) > 0.$$

The value at $t = 0$ is independent of e , so K_B is too. Formula (7) remains valid when some differences $e_b - e_{a+1}$ vanish; it does not assert an exact order of vanishing in those directions.

Let $r = \min_{B \in \mathcal{S}} n_I(B)$. If $r < m$, choose e_k strictly decreasing on T . Every $P_{B_I}(e)$ is then positive. Apply $\psi_{t,e}$ to (3), divide by t^r , and let $t \rightarrow 0$. The left side tends to zero, whereas the right side tends to $\sum_{n_I(B)=r} c_B K_B P_{B_I}(e) > 0$. This proves (4).

Now take arbitrary e , divide the specialized identity by t^m , and let $t \rightarrow 0$. We get (5) evaluated at e . Since this holds for every $e \in \mathbb{R}^T$, it is a polynomial identity. $\square$

The count inequality holds for *every* interval in one and the same decomposition. The constants in (5) may change with the interval; no comparison between constants from different intervals is required.

COROLLARY 3.2 (Insertion of a rectangular pair). *If s_D spans an extreme ray, then so does $s_{D \cup \{(p,p)\}}$ for every $p \geq 1$.*

Proof. In a positive decomposition of $s_D s_{(p,p)}$, apply (4) with $I = [p, p]$. Every term contains (p, p) . Cancel one copy of its Schur factor in the polynomial ring. A decomposition using other multisets would give a decomposition of s_D using other multisets, contrary to extremality and Lemma 2.1. $\square$

4. ENDPOINTS AND THEIR MONOMIALS

For a multiset M of m pairs, let $\tau(M)$ and $\beta(M)$ be its multisets of tops and bottoms. List them increasingly as $a_1(M) \leq \dots \leq a_m(M)$ and $b_1(M) \leq \dots \leq b_m(M)$. Write $\text{wt}(M) = \sum_i a_i(M) + \sum_i b_i(M)$ and

$$P_M = \prod_{(a,b) \in M} (\delta_b - \delta_{a+1}), \quad \delta^X = \prod_{v \in X} \delta_v.$$

In $\delta^{\tau(M)+1}$, the 1 is added to every member of the multiset $\tau(M)$.

LEMMA 4.1 (Endpoint inequalities). *Let D, M consist of m pairs inside $[w, u]$. If $n_J(M) \geq n_J(D)$ for every subinterval $J \subseteq [w, u]$, then*

$$(8) \quad b_i(M) \geq b_i(D), \quad a_i(M) \leq a_i(D) \quad (1 \leq i \leq m).$$

Proof. The intervals $[x, u]$ count bottoms at least x , and the intervals $[w, y]$ count tops at most y . Their inequalities are precisely the two coordinatewise comparisons between the increasing lists. $\square$

LEMMA 4.2 (Endpoint monomials). *Let D, M consist of m pairs.*

(1) *If $\sum \beta(M) \geq \sum \beta(D)$, then*

$$[\delta^{\beta(D)}] P_M = \begin{cases} 1 & \text{if } \beta(M) = \beta(D), \\ 0 & \text{otherwise.} \end{cases}$$

(2) *If $\sum \tau(M) \leq \sum \tau(D)$, then*

$$[\delta^{\tau(D)+1}] P_M = \begin{cases} (-1)^m & \text{if } \tau(M) = \tau(D), \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Give δ_k weight k . In expanding P_M , choosing δ_{a+1} instead of δ_b strictly increases the weight, since $a+1 > b$. Thus the unique choice of least weight selects every bottom; its monomial is $\delta^{\beta(M)}$, with coefficient 1. Under the first hypothesis, the target has weight no greater than this minimum, so it can occur only as that term. The unique choice of greatest weight selects every top plus one and has coefficient $(-1)^m$. The second assertion follows in the same way. Repeated factors cause no difficulty, since every change of choice changes the weight strictly. $\square$

LEMMA 4.3 (Reconstruction). *A nested multiset of pairs is determined by its tops and its bottoms.*

Proof. Let b be the largest bottom, with multiplicity r . We claim that the r tops paired with b are the r smallest available tops at least b . Indeed, let a^* be the largest top paired with b . If a pair (a, d) has $d < b \leq a < a^*$, then (a^*, b) and (a, d) interlace. Thus every top at least b that is paired with a smaller bottom is at least a^* . This proves the claim, including ties. Remove the r pairs just determined, and repeat. Induction reconstructs the entire multiset. $\square$

5. PROOF OF THE MAIN THEOREM

White's theorem proves the non-nested direction. Suppose A is nested and s_A is not extreme. Lemma 2.2 gives a decomposition (3) with nested $B \neq A$ and $\varphi(B) \geq \lambda := \varphi(A)$.

Let D be the multiset of pairs in A , put $m = |D|$, and let $n = \ell(\lambda)$. Nestedness gives $n \in \{2m, 2m + 1\}$. Let w and u be the smallest and largest parts of λ , and put $I = [w, u]$. Dominance implies $\ell(\varphi(B)) \leq n$: a partition with more than n positive parts cannot dominate a partition of the same size with n parts. Hence the interval lemma gives

$$(9) \quad 2m \leq 2n_I(B) \leq \ell(\varphi(B)) \leq n \leq 2m + 1.$$

It follows that B has exactly m pairs, all inside I , and at most one singleton. Denote its pair multiset by M_B , and write

$$c'_B = N - \text{wt}(M_B) \geq 0.$$

A positive c'_B is the singleton of B ; $c'_B = 0$ means that it has none. If A has no singleton, (9) forces $c'_B = 0$.

For every subinterval $J \subseteq I$, Lemma 3.1 gives

$$(10) \quad n_I(M_B) \geq n_I(D).$$

Its polynomial identity for I , after division by K_A , is

$$(11) \quad P_D = \sum_{B \in \mathcal{S}} \alpha_B P_{M_B}, \quad \alpha_B = c_B K_B / K_A > 0.$$

We show that a particular coefficient on the left cannot be supplied by any term on the right.

5.1. No singleton. By Lemma 4.1, the bottoms of M_B are coordinatewise at least those of D , and its tops are coordinatewise at most those of D . Extract the coefficient of $\delta^{\beta(D)}$ in (11). It is 1 on the left. Lemma 4.2 says that a term on the right can contribute only if $\beta(M_B) = \beta(D)$. For such a term, equality of total weights gives

$$\sum \tau(M_B) = N - \sum \beta(M_B) = N - \sum \beta(D) = \sum \tau(D).$$

The coordinatewise top inequalities must therefore all be equalities. Reconstruction gives $M_B = D$, so $B = A$, a contradiction.

5.2. A singleton. Suppose $A = D \uplus \{(c)\}$. Nestedness gives a disjoint splitting

$$D = D_- \uplus D_0 \uplus D_+,$$

where

$$D_- = \{(a, b) \in D : a < c\}, \quad D_0 = \{(c, c) \in D\}, \quad D_+ = \{(a, b) \in D : b > c\}.$$

Write $m_- = |D_-|$, $m_0 = |D_0|$, and $m_+ = |D_+|$. Indeed, every nonrectangular pair avoids c , and every rectangular pair belongs to exactly one of these classes.

Apply (10) to $[w, c-1]$, $[c, c]$, and $[c+1, u]$, omitting empty intervals. Each M_B has at least m_- , m_0 , and m_+ pairs in these three disjoint classes. Since their sum is m , all three counts are exact and there are no other pairs. Thus

$$(12) \quad M_B = M_- \uplus D_0 \uplus M_+, \quad |M_-| = m_-, \quad |M_+| = m_+.$$

The notation M_- and M_+ depends on B . Applying the endpoint lemma within each of the lower and upper intervals gives

$$(13) \quad \begin{aligned} b_i(M_-) &\geq b_i(D_-), & a_i(M_-) &\leq a_i(D_-), \\ b_i(M_+) &\geq b_i(D_+), & a_i(M_+) &\leq a_i(D_+). \end{aligned}$$

Cancel the common nonzero polynomial $P_{D_0} = (\delta_c - \delta_{c+1})^{m_0}$ in (11). This leaves

$$(14) \quad P_{D_-} P_{D_+} = \sum_{B \in \mathcal{S}} \alpha_B P_{M_-} P_{M_+}.$$

Consider the monomial

$$(15) \quad \theta = \delta^{\tau(D_-)+1} \delta^{\beta(D_+)}.$$

The lower polynomials use only variables with index at most c ; the upper polynomials use only variables with index at least $c+1$. Their variable sets are disjoint. By Lemma 4.2, the coefficient of θ on the left is $(-1)^{m_-}$, and a term on the right has this same coefficient precisely when

$$(16) \quad \tau(M_-) = \tau(D_-), \quad \beta(M_+) = \beta(D_+);$$

otherwise its coefficient is zero. Consequently at least one $B \in \mathcal{S}$ satisfies (16).

We finish by showing that such a B must be A . Define

$$p = \sum \beta(M_-) - \sum \beta(D_-) \geq 0, \quad q = \sum \tau(D_+) - \sum \tau(M_+) \geq 0.$$

The nonnegativity follows from (13). Equality $p = 0$ forces equality of the bottom multisets in the lower block; equality $q = 0$ forces equality of the top multisets in the upper block. By (16),

$$(17) \quad \text{wt}(M_-) = \text{wt}(D_-) + p, \quad \text{wt}(M_+) = \text{wt}(D_+) - q, \quad c'_B = c - p + q.$$

For the dominance comparisons, include c'_B as a part even when it is zero. The resulting multiset Π_B has exactly n entries and the same total weight as λ . Dominance says that the sum of its k largest entries is at least the corresponding sum for λ ; equivalently, its k smallest entries have sum at most the corresponding sum for λ .

The upper block forces $q = 0$. First suppose $c'_B > c$. If $m_+ = 0$, then $q = 0$ and $c'_B = c - p \leq c$, a contradiction. Otherwise put $r = 2m_+ > 0$. The r parts of M_+ and the entry c'_B are all at least $c+1$; every other entry of Π_B is at most c . Its r largest entries therefore have sum at most

$$\text{wt}(M_+) + c'_B - (c+1) = \text{wt}(D_+) - p - 1 < \text{wt}(D_+).$$

But the r largest parts of λ are exactly the parts of D_+ . This contradicts dominance. Thus $c'_B \leq c$. Now the r largest entries of Π_B are the parts of M_+ , whose sum is $\text{wt}(D_+) - q$. Dominance forces $q = 0$. This conclusion also holds when $r = 0$, since then both upper blocks are empty. We have proved $c'_B = c - p \leq c$.

The lower block forces $p = 0$. If $m_- = 0$, this is immediate. Otherwise put $s = 2m_- > 0$ and let

$$z = \max(\{\text{parts of } M_-\} \cup \{c'_B\}).$$

These $s + 1$ entries are at most c , while every other entry of Π_B is at least c . Its s smallest entries have sum

$$\text{wt}(M_-) + c'_B - z = \text{wt}(D_-) + c - z.$$

The s smallest parts of λ have sum $\text{wt}(D_-)$. Dominance therefore implies $z \geq c$. Since every part of M_- is strictly less than c and $c'_B \leq c$, we obtain $c'_B = c$ and $p = 0$. Ties at c , including those in D_0 , do not change the displayed sum.

All endpoint inequalities in (13) are now equalities, using (16). The multisets are nested, so Lemma 4.3 gives $M_- = D_-$ and $M_+ = D_+$. Together with (12) and $c'_B = c$, this yields $B = A$, the required contradiction. This completes the proof of Theorem 1.1. $\square$

6. FURTHER REMARKS

The proof uses Schur positivity and dominance to restrict a hypothetical decomposition, and then works entirely with Jacobi–Trudi factors. The interval lemma requires no nestedness. Nestedness enters in the bound of one singleton, the splitting around that singleton, and the reconstruction of pairs from their endpoints. Dominance supplies the length bound and the two comparisons at the end of the proof.

The last comparisons cannot simply be omitted. For example,

$$A = \{(5, 4), (3), (2, 1)\}, \quad B = \{(4, 4), (4), (2, 1)\}$$

are nested, satisfy the interval count inequalities and the endpoint conditions (16), but $\varphi(B)$ does not dominate $\varphi(A)$.

The argument rules out every positive decomposition and therefore implies the existence of separating linear functionals by finite-dimensional separation. It does not supply an explicit formula for one, or prove the support restriction in White's Conjecture 11 on $[\lambda, \lambda^{++}]$ [1]. For fixed e , Taylor coefficients of $\psi_{t,e}$ are linear functionals on Λ_N ; understanding their Schur supports may help relate the present proof to that stronger question. The argument also gives no classification for factors with three or more parts: the two-part factorization (6) is essential here.

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